Vassily Lombard

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Education

Columbia University **Sept. 2025 – May 2027**
BS in Computer Engineering - Richard and Janet Hunter Scholar & Dean's List GPA: 3.79 – New York, NY
Relevant Coursework: Analog Electronics, VLSI Circuits, Logic Design, Photonics, Solid State Device

Earlham College **Sept. 2022 – May 2025**
BA in Data Science - William Roha Mathematics Award Recipient GPA: 3.97 – Richmond, IN
Relevant Coursework: Artificial Intelligence, Database Management, Data Modeling, Stats & Probability

Experience/Leadership

R&D Pixel Design Intern **May 2026 - Sept. 2026**
STMicroelectronics Crolles, France
- Designed mixed-signal sub-blocks (ADC, sampling stages, and readout chains) for a global shutter pixel array, performing Spectre simulations to optimize circuit specs and ensure full-system functionality.
- Re-architectured the SRAM write-path in Cadence, optimizing transistor sizing and parasitic capacitance to achieve a 2x increase in maximum clock frequency.
- Compiled custom hardware testbenches to characterize physical silicon performance in a cleanroom environment, measuring key metrics including dark current and quantum efficiency.
- Achieved the primary goal of the internship by capturing the sensor's first darkroom image, confirming the innovative aspect of the testchip.

Research Assistant **Nov. 2025 - May 2026**
Columbia University UEIL Lab New York City, NY
- Developed quantitative metrics to analyze ultrasound localization microscopy brain images.
- Engineered a bubble tracking algorithm to reconstruct a map of brain vessels accurate at 86%.

Machine Learning Engineer Intern **May 2025 – Aug. 2025**
Palo IT Paris, France
- Optimized the MoE expert selection architecture in Mixtral, improving load balancing across experts by 9%.
- Implemented a local ChatBot with MCP servers to support different skills reducing documentation writing time by 30%.
- Conceived a RAG based app with AWS Lambda and API Gateway to enhance LLM answers for technical documentation.

Robotics Club President **Sept. 2024 – May 2025**
Earlham College Richmond, IN
- Developed an autonomous mobile robot in C on Arduino, integrating light and proximity sensors with motor control for real-time navigation.
- Implemented sensor-based line following and obstacle detection, enabling the robot to adapt its trajectory in real time.
- Led a 20+ member robotics club, organizing projects and an end-of-semester autonomous robot competition.

Projects

2.4-GHz CMOS LNA for Receiver | *Cadence Virtuoso, SpectreRF, 130-nm CMOS PDK* **Sept. 2026 – Now**
- Design a 2.4-GHz inductively degenerated common-source LNA for a 50-Ω receiver front-end, optimizing transistor sizing, biasing, source degeneration, and LC input/output matching.
- Optimize RF performance across gain, noise figure, impedance matching, and power consumption, targeting > 12 dB gain, < 3.5 dB noise figure, and < 10 mW power consumption.
- Developing the RFIC layout with on-chip inductors and matching networks, including DRC/LVS, parasitic extraction, and post-layout S-parameter and noise analysis.

BiCMOS Bandgap Voltage Reference | *Cadence Virtuoso, Spectre, 130-nm CMOS PDK* **Jul. 2026 - Aug. 2026**
- Designed a transistor-level ∼1.2-V bandgap voltage reference using PTAT/CTAT temperature compensation, bipolar devices, current mirrors, and feedback circuitry in Cadence Virtuoso.
- Characterized voltage reference performance across temperature, PVT corners, PSRR, noise, and power, and evaluated process variation and mismatch using Monte Carlo analysis.
- Completed physical IC layout and verified DRC/LVS, followed by parasitic extraction and pre- vs. post-layout performance comparison.

Skills/Interests

Languages: Python, Java, C/C++, Matlab, SQL, Verilog, VHDL
Developer Tools: Cadence Virtuoso, Spectre, AWS Cloud, Microsoft Azure (Certified AI-900), GitHub
Libraires: PyTorch, TensorFlow, OpenCV, FastAPI
Activities and Interests: Competitive golf (Scratch handicap), Strength training (315 lb bench)